

BEYOND STANDALONE PERFORMANCE: WHAT DO TIME-SERIES ANOMALY DETECTORS ADD BEYOND SIMPLE STATISTICS?

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ABSTRACT

Sophisticated time-series anomaly detectors do not always outperform simple statistical scorers. What do their scores add beyond those statistics? We introduce conditional detection utility (CDU), the held-out log-loss reduction from adding a detector score to a fixed statistical reference. On 350 univariate series from 23 sources, we evaluate fixed scores from nine detectors with source-held-out probes. Standalone performance and conditional contribution give different rankings even under the same probe and loss. MOMENT-ZS scores are highly reconstructible from the reference ($R^2 = 0.904$) yet rank ninth by CDU. The leading conditional ranks persist across probe families, sampling seeds, and reference variants. CDU provides a complementary evaluation axis for heterogeneous TSAD scores.

Index Terms— Time-series anomaly detection, conditional utility, statistical reference, source-held-out evaluation

1. INTRODUCTION

Modern time-series anomaly detection (TSAD) uses increasingly sophisticated detectors, yet simple scorers based on local variation and deviation remain surprisingly competitive [1]. When a complex detector performs well, does it capture anomaly-relevant structure beyond simple statistics, or largely re-express patterns already present in the time series? Benchmarks such as TSB-AD [2] characterize standalone detection performance but do not directly answer what a detector adds beyond a statistical reference.

Work on foundation models asks whether learned scores provide value beyond window statistics [3]. Predictive $\mathcal{V}$ -information studies what a restricted predictor can access [4]; conditional probing tests what one representation adds beyond another [5]; and predictive variable importance compares risks with and without a feature set [6]. These ideas suggest examining detector scores relative to a fixed statistical reference under source-held-out transfer.

A detector score S_D can be viewed from three perspectives. First, *standalone performance* measures how well it identifies anomalies, for example with VUS-PR [7]. Second, *statistical*

overlap measures how well a reference B predicts S_D . Finally, *conditional contribution* asks whether S_D improves label prediction once B is available. A highly predictable score can retain a useful residual; conversely, a score unlike B may contain only irrelevant noise.

We measure that contribution with conditional detection utility (CDU): the reduction in held-out log-loss when a detector score is added to the same reference. In our nine-detector comparison, standalone and conditional ranks differ under the same probe and loss; score reconstructibility from the reference does not predict CDU ($\rho = -0.167$). Code and result evidence are available at <https://github.com/echo8383/CDU>.

2. CONDITIONAL DETECTION UTILITY

2.1. Conditional detection utility

Let $Y \in \{0, 1\}$ denote an anomaly label, $B \in \mathbb{R}^p$ a statistical reference, and S_D a detector score. For each held-out source, fit two probability predictors from the same probe family $\mathcal{Q}$: one receives B , and the other (B, S_D) . Both use the same training points, weights, and fitting settings. The operational estimate is

$$\widehat{\text{CDU}}_{\mathcal{Q}}(D | B) = \widehat{L}_B - \widehat{L}_{BD}, \quad (1)$$

where $\widehat{L}_B$ and $\widehat{L}_{BD}$ are their paired held-out log-losses in bits. Positive CDU means that the detector improves label prediction given the reference. Figure 1 summarizes the comparison.

For a matched standalone comparison, $U_D = \widehat{L}_0 - \widehat{L}_D$, where the detector-only probe receives S_D and the null uses training-source anomaly prevalence. Thus U_D and CDU use the same loss but different reference information.

Define $\mathcal{R}^*(X) = \inf_q \mathbb{E}[-\log_2 q(Y | X)]$, where q ranges over conditional probability distributions.

Proposition 1. Under a common joint distribution and unrestricted predictors,

$$\begin{aligned} \text{CDU}^*(D | B) &= \mathcal{R}^*(B) - \mathcal{R}^*(B, S_D) \\ &= I(Y; S_D | B). \end{aligned} \quad (2)$$

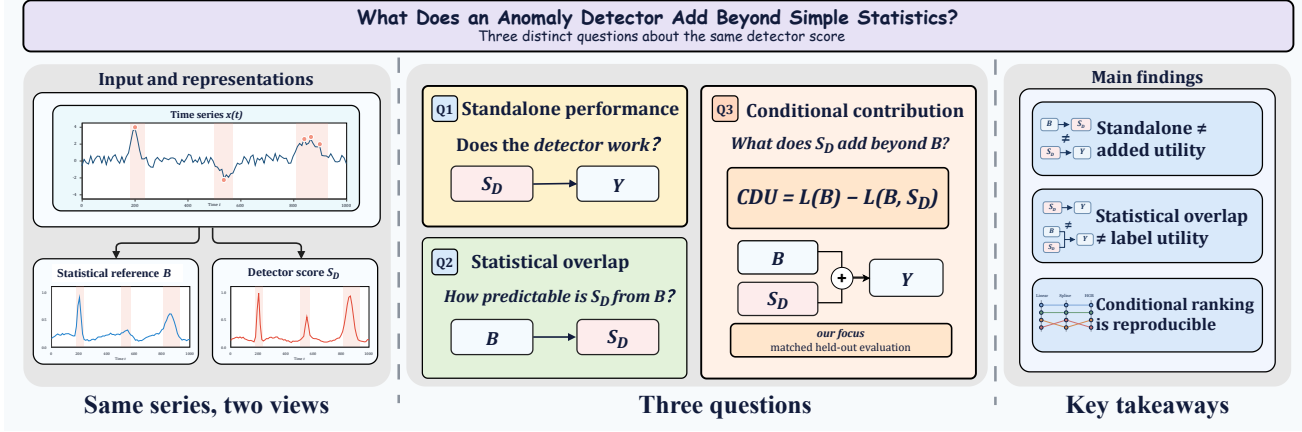


Fig. 1. The same series yields a statistical reference and a detector score. We distinguish standalone performance (Q1), statistical overlap (Q2), and conditional contribution (Q3); the three comparisons answer different questions.

Proof. For any input X , the cross-entropy decomposition gives

$$\mathbb{E}[-\log_2 q(Y | X)] = H(Y | X) + \mathbb{E}_X D_{\text{KL}}(P_{Y|X} \| q_{Y|X}). \quad (3)$$

The KL term is nonnegative and vanishes for the true conditional distribution, so $\mathcal{R}^*(X) = H(Y | X)$. Therefore,

$$\mathcal{R}^*(B) - \mathcal{R}^*(B, S_D) = H(Y | B) - H(Y | B, S_D) = I(Y; S_D | B). \quad (4)$$

This proves the identity. $\square$

Population CDU is nonnegative and vanishes when $Y \perp S_D | B$, including $S_D = g(B)$. Finite-probe CDU is predictor-family-relative and may be negative under transfer; it is not an exact estimate of conditional mutual information.

2.2. Score interface

Each curve is aligned to the same time indices as its labels. We evaluate within-series anomaly ordering through average ranks, removing detector-specific score spacing. For a length- T_i curve, ranks start at one:

$$\tilde{s}_{it} = \frac{\text{rank}_{\text{avg}}(s_{it}) - 0.5}{T_i}. \quad (5)$$

The statistical reference uses the same rank representation. Ties are preserved, and constant curves map to 0.5. Rank normalization is part of the evaluation definition: it measures the label information added by a detector's anomaly ordering beyond the rank-normalized statistical reference. The transform is label-free; B and S_D below denote these probe inputs.

2.3. Source-held-out estimation

A source is a benchmark dataset group. Each outer fold excludes all series of one source from probe training. La-

bels from the remaining sources supervise the probes; held-out labels are used only to compute losses. Detector outputs and the statistical reference remain fixed. This strict split prevents probes from exploiting sibling series from the held-out source.

Let G_{tr} be the number of training sources and n_g the number of series in training source g . If m_i points are retained from series i and $N_s = \sum_i m_i$, assign each retained point weight

$$w_{it} = \frac{N_s}{G_{\text{tr}} n_g m_i}. \quad (6)$$

This gives equal total weight to sources and to series within each source. Training retains $m_i = \min(T_i, 2048)$ points per series; every test point is evaluated. A common basis-only predictor supplies the reference loss for all detectors within each fold. With binary log-loss ℓ and held-out probability p_{it}^M for model M , aggregation is

$$L_i^M = \frac{1}{T_i} \sum_{t=1}^{T_i} \ell(y_{it}, p_{it}^M), \quad (7)$$

$$\hat{L}_M = \frac{1}{G} \sum_{g=1}^G \frac{1}{n_g} \sum_{i \in g} L_i^M.$$

Here $G = 23$; paired source-bootstrap intervals quantify variation across held-out sources.

3. EXPERIMENTAL SETUP

Data and detectors. We evaluate 350 univariate series from TSB-AD-U [2], following the evaluation population in [1]. Source groups follow the dataset identifier in each filename; MSL and SMAP are distinct, and UCR is grouped as one archive. We use SubPCA and POLY benchmark implementations [2], fine-tuned (FT) and zero-shot (ZS)

Table 1. Primary spline results for nine fixed detector scores. VUS-PR is series-macro; U_D and CDU are in bits. Parentheses give ranks; brackets give source-bootstrap 95% intervals.

Detector	VUS-PR (rank)	U_D (rank)	CDU [95% CI]	CDU rank
SubPCA	0.4224 (1)	0.04060 (1)	0.01509 [0.00256, 0.02964]	1
POLY	0.3893 (2)	0.02700 (3)	0.00604 [0.00246, 0.00969]	3
MOMENT-FT	0.3864 (3)	0.01907 (4)	0.00099 [-0.00001, 0.00208]	7
MOMENT-ZS	0.3832 (4)	0.01832 (5)	0.00011 [-0.00088, 0.00113]	9
M2N2	0.2870 (5)	0.03090 (2)	0.01272 [0.00075, 0.02614]	2
TranAD	0.2754 (6)	0.01438 (6)	0.00484 [-0.00162, 0.01263]	4
TimesNet	0.2627 (7)	0.00591 (8)	0.00133 [-0.00015, 0.00300]	5
FITS	0.2464 (8)	0.00598 (7)	0.00109 [-0.00008, 0.00221]	6
AnomalyTransformer	0.1118 (9)	-0.00005 (9)	0.00015 [-0.00129, 0.00167]	8

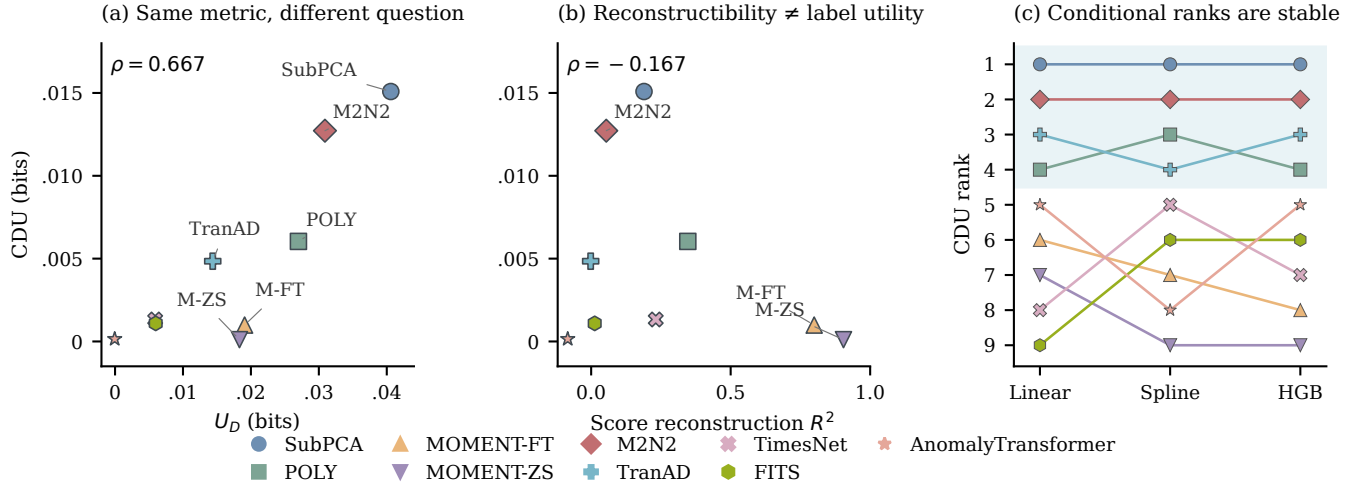


Fig. 2. The benchmark blind spot and its consequences. (a) Detector-only and conditional utility use the same spline probe and log-loss yet differ; (b) statistical reconstructibility does not determine CDU; (c) the top-two order and top-four membership (shaded) persist across probes.

MOMENT [8], M2N2 [9], TranAD [10], TimesNet [11], FITS [12], and AnomalyTransformer [13]. All 350 curves are retained for every detector.

Statistical reference. The 31 fixed, label-free features span local variance, range, prediction deviation, absolute difference, robust dispersion, and spectral entropy. No feature is selected using held-out labels.

Probes and comparisons. Additive cubic-spline logistic regression is the primary probe; linear logistic and histogram gradient boosting (HGB) test probe sensitivity. All use the same label-independent training sample (at most 2048 points per series), source/series weights, and full test curves. Logistic probes use $C = 0.1$, spline has four fixed knots, and HGB runs for 32 iterations. VUS-PR and CDU use the same detector curves.

Reconstruction and robustness. Source-held-out ridge regression predicts ranked detector scores from B ($\alpha = 1$); we report source-macro R^2 and Spearman correlation. We repeat the spline comparison across five sampling seeds, seven

reference-family removals, cumulative 7–29-feature references, and alternative detector-score interfaces (min–max or Gaussianized z-scores).

Controls. We test a duplicated reference feature, independent Gaussian noise, and a synthetic score $Z + 2Y$ with independent $Z \sim \mathcal{N}(0, 1)$. Only the last control uses labels.

4. RESULTS AND INTERPRETATION

4.1. Standalone and conditional rankings differ

Table 1 and Fig. 2a compare detector-only utility U_D with CDU under the same spline probe, log-loss, held-out sources, and weights. MOMENT-FT and MOMENT-ZS fall from U_D ranks 4 and 5 to CDU ranks 7 and 9, while TranAD rises from 6 to 4. Across linear, spline, and HGB probes, the U_D /CDU rank correlations are 0.967, 0.667, and 0.500. M2N2 and TranAD move from VUS ranks 5/6 to CDU ranks 2/4.

POLY and both MOMENT variants have similar VUS-PR (0.3832–0.3893), yet POLY exceeds MOMENT-FT by 0.00505 bits of spline CDU [0.00172, 0.00860] and MOMENT-ZS by 0.00593 [0.00241, 0.00961]. Both differences remain positive after a joint max-standardized bootstrap correction over three pairwise POLY/MOMENT comparisons across the three probes.

4.2. Score reconstructibility does not determine CDU

Source-held-out ridge regression reconstructs MOMENT-FT and MOMENT-ZS scores strongly (source-macro $R^2 = 0.799$ and 0.904 ; Spearman 0.895 and 0.951), but M2N2 and TranAD weakly ($R^2 = 0.054$ and -0.002). Across nine detectors, reconstruction R^2 and spline CDU have Spearman correlation -0.167 (Fig. 2b). Low reconstructibility alone does not imply conditional contribution; residual score variation matters only when it improves label prediction.

4.3. Conditional rankings are stable

SubPCA and M2N2 remain first and second across linear, spline, and HGB; TranAD and POLY complete the same top-four set (Fig. 2c). Pairwise rank correlations are 0.717 – 0.867 . Spline and HGB achieve lower basis-only loss than linear logistic (0.2891 and 0.2869 vs. 0.3375 bits); giving the linear probe all training points barely changes its loss (0.337462 versus 0.337447).

Five training-sampling seeds give identical spline ranks. Seven reference-family removals give rank correlations of 0.917 – 1.000 ; cumulative 7–29-feature references retain the top four ($\rho = 0.933$ – 1.000). Gaussianized z-scores also retain that group ($\rho = 0.933$). Min-max scaling retains detector-specific score spacing and changes broader ranks ($\rho = 0.517$), moving POLY from rank 3 to 9. The primary rank interface therefore evaluates anomaly ordering rather than detector-specific score geometry.

4.4. Controls separate redundancy from label signal

Under the primary spline probe, duplicate and independent-noise controls remain near zero, whereas the label-informed synthetic score gives positive CDU (0.09220 bits) in all 23 sources. Thus the paired comparison responds to anomaly-relevant information rather than mere feature addition.

5. DISCUSSION AND CONCLUSION

Complex TSAD detectors can perform well while overlapping strongly with simple statistical structure. Standalone performance, statistical overlap, and conditional contribution give different views of the same score; CDU quantifies the third. Across nine detectors, the conditional ranking differs from standalone evaluation and remains broadly consistent across probes, seeds, and reference variants. Reporting these

views together clarifies what a detector adds beyond the fixed statistics.

Such comparisons may inform detector selection: models with similar standalone performance need not be interchangeable once their conditional contributions are measured. Using reference-relative utility to guide model combination or representation learning remains future work.

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